

FOOD DELIVERY PERFORMANCE ANALYSIS

Project Documentation

Project Title

Food Delivery Data Analysis

Project Type

Data Analytics Project

Tools & Technologies

Python | Pandas | NumPy | Seaborn | Matplotlib | MySQL |

Power BI | Jupyter Notebook

1. PROJECT OVERVIEW

The Food Delivery Data Analysis project focuses on analyzing food delivery operations to understand the factors that influence delivery time and overall delivery performance.

The project uses a food delivery dataset containing 45,584 records and 20 columns. The data includes information related to delivery time, delivery person details, ratings, traffic conditions, weather conditions, vehicle type, order type, city, festival conditions, and multiple deliveries.

The project follows an end-to-end data analytics workflow, starting with data cleaning and preparation using Python and Pandas, followed by exploratory and business analysis using Python and SQL. The final insights are presented through an interactive Power BI dashboard.

2. PROJECT OBJECTIVES

The primary objective of this Food Delivery Data Analysis project is to analyze food delivery data and identify the major factors that influence delivery time and delivery performance.

The project aims to clean and prepare the raw dataset using Python and Pandas so that the data can be used reliably for further analysis. Data quality issues such as invalid values, duplicate records, missing values, inconsistent text values, and data validation problems are identified and handled during the cleaning process.

Another objective is to perform exploratory data analysis to understand the distribution and characteristics of the food delivery data. Different delivery-related factors such as traffic conditions, weather conditions, vehicle type, order type, city, festival conditions, and multiple deliveries are analyzed to identify meaningful patterns.

The project also aims to use SQL to perform business-oriented analysis on the cleaned dataset. SQL queries are used to calculate and compare average delivery times across different

categories and to answer important business questions related to delivery performance.

Finally, the project aims to present the major findings through an interactive Power BI dashboard. The dashboard provides key performance indicators, business visualizations, and interactive slicers that help users understand delivery performance and explore the data from different perspectives.

Overall, the project aims to demonstrate how raw business data can be transformed into meaningful insights using a complete data analytics workflow involving Python, Pandas, SQL, data visualization, and Power BI.

3. DATASET DESCRIPTION

The dataset used in this project contains food delivery records collected for analyzing delivery operations and performance.

The dataset consists of 45,584 records and 20 columns, providing information about different aspects of the food delivery process.

The dataset contains information related to delivery persons, their ratings, vehicle details, weather conditions, road traffic density, order types, city classification, festival conditions, multiple deliveries, and delivery time. These attributes provide useful information for understanding the operational factors that may influence the time taken to complete a delivery.

Before performing the analysis, the raw dataset was carefully inspected and validated using Python and Pandas. The data was then cleaned to improve its quality and consistency. The cleaned dataset was saved separately and later used for SQL analysis and Power BI dashboard development.

The important categories of data used in the analysis include Weather Conditions, Road Traffic Density, Type of Vehicle,

**Type of Order, City, Festival, Multiple Deliveries, Rating, and
Delivery Time.**

**The cleaned dataset was maintained as a separate CSV file
named food_delivery_cleaned.csv and was used as the main
dataset for the subsequent analysis stages.**

4. TOOLS AND TECHNOLOGIES

This project was developed using a combination of programming, database, data visualization, and business intelligence tools. Each tool was used for a specific stage of the data analytics workflow.

Python was used as the primary programming language for data analysis and data processing. It provided the environment required to inspect, clean, transform, and analyze the food delivery dataset.

Pandas was used for data cleaning, data manipulation, validation, and exploratory analysis. It was mainly used to identify data quality issues and prepare the dataset for further analysis.

NumPy was used for numerical operations and supporting data processing tasks during the analysis.

Seaborn and Matplotlib were used to create data visualizations such as histograms and bar plots. These visualizations helped identify patterns and relationships within the delivery data.

MySQL was used as the database platform for storing the cleaned dataset and performing SQL-based business analysis.

SQL queries were used to answer business questions and compare delivery performance across different categories.

Power BI was used to develop the final interactive dashboard. The dashboard contains KPI cards, business visuals, and slicers that allow users to explore delivery performance interactively.

Jupyter Notebook was used as the working environment for Python-based data cleaning, analysis, and visualization.

Together, these technologies were used to create a complete end-to-end data analytics workflow from raw data preparation to business insights and interactive dashboard reporting.

5. PROJECT WORKFLOW

The Food Delivery Data Analysis project follows a structured end-to-end data analytics workflow. The workflow begins with the raw dataset and continues through data cleaning, exploratory analysis, SQL business analysis, visualization, and dashboard development.

5.1 Raw Data Collection

The original food delivery dataset was used as the starting point of the project. The dataset contained 45,584 records and 20 columns representing different aspects of food delivery operations.

5.2 Data Cleaning and Preparation

The raw dataset was imported into Python using Pandas. The data was inspected for missing values, duplicate records, invalid values, inconsistent text values, and data quality issues. The identified issues were handled and the cleaned dataset was validated.

5.3 Exploratory Data Analysis

After cleaning, exploratory data analysis was performed to understand the characteristics and patterns of the dataset. Statistical summaries and categorical analysis were used to understand delivery performance across different factors.

5.4 SQL Business Analysis

The cleaned dataset was imported into MySQL. SQL queries were then used to perform business analysis and answer questions related to average delivery time across traffic, weather, vehicle, order type, city, festival conditions, and multiple deliveries.

5.5 Data Visualization

Python visualization libraries such as Seaborn and Matplotlib were used to create charts that helped identify important patterns in delivery performance.

5.6 Power BI Dashboard

The analyzed data was finally presented through an interactive Power BI dashboard. The dashboard includes KPI

cards, business visuals, and slicers that allow users to explore the delivery data interactively.

5.7 Final Business Insights

The results from the analysis were interpreted to identify important factors affecting delivery performance and to provide meaningful business insights.

Overall, the workflow transforms raw food delivery data into a cleaned, analyzed, visualized, and business-ready analytical solution.

6. DATA CLEANING AND VALIDATION

The raw food delivery dataset was cleaned and validated using Python and Pandas before performing further analysis. Data cleaning was an important step because the quality and consistency of the dataset directly affect the accuracy of the analysis and business insights.

The dataset was first inspected to understand its structure, number of records, columns, data types, and overall data quality. Missing values and duplicate records were checked to identify any incomplete or repeated records.

The rating column was carefully validated to identify invalid rating values. The identified invalid ratings greater than the valid rating range were replaced using the median value of the rating column.

The dataset was also checked for invalid values in important operational columns. Multiple delivery values were verified to ensure that they remained within the expected range. Delivery time values were checked to ensure that there were no invalid non-positive values.

String columns were examined for unnecessary whitespace and inconsistent text formatting. Leading and trailing whitespaces were removed wherever required to maintain consistent categorical values.

The dataset was also validated after the cleaning process. Missing value checks and duplicate checks were performed again to ensure that the cleaned dataset was ready for further analysis.

After completing the cleaning and validation process, the final dataset contained 45,584 records and 20 columns. The cleaned dataset was exported as food_delivery_cleaned.csv and used for the subsequent SQL analysis and Power BI dashboard development.

The data cleaning process ensured that the dataset was consistent, reliable, and suitable for exploratory analysis, business analysis, and visualization.

7. EXPLORATORY DATA ANALYSIS

After completing the data cleaning and validation process, Exploratory Data Analysis (EDA) was performed to understand the characteristics, distribution, and patterns present in the food delivery dataset.

The analysis focused mainly on delivery time and the different operational factors that may influence delivery performance. Descriptive statistics were used to understand the numerical variables, while categorical analysis was performed to understand the distribution of different delivery-related categories.

The distribution of delivery time was analyzed to identify the typical time taken for food deliveries. The analysis showed that most deliveries were completed within the lower delivery-time range, while a smaller number of deliveries took considerably longer.

Different traffic density categories were analyzed to understand their relationship with delivery time. Weather conditions were also examined to identify whether adverse

weather conditions were associated with longer delivery durations.

Vehicle types were compared based on their average delivery time to understand differences in delivery performance.

Similarly, different order types were analyzed to determine whether the type of food order created noticeable differences in delivery duration.

City categories, festival conditions, and multiple delivery situations were also explored to identify additional patterns in delivery performance.

The EDA results provided an initial understanding of the dataset and helped identify the major areas for further business analysis using SQL and data visualization.

Overall, the exploratory analysis helped transform the cleaned dataset into meaningful patterns that could be investigated further through business-oriented questions.

8. SQL BUSINESS ANALYSIS

After completing the data cleaning and exploratory analysis, the cleaned food delivery dataset was imported into MySQL for business-oriented analysis.

SQL was used to analyze the dataset from a business perspective and to answer important questions related to delivery performance. The analysis mainly focused on understanding how different operational factors influence the average delivery time.

A total of 17 business analysis questions were performed using SQL. The analysis used concepts such as SELECT, WHERE, ORDER BY, LIMIT, DISTINCT, LIKE, IN, BETWEEN, Aggregate Functions, GROUP BY, HAVING, JOIN, and SUBQUERY.

The major areas analyzed were road traffic density, weather conditions, vehicle type, order type, city, festival conditions, and multiple deliveries.

The analysis compared the number of orders and average delivery time across different categories. This helped identify categories associated with higher or lower delivery times and

provided a business-oriented understanding of delivery performance.

The SQL analysis also served as a bridge between the cleaned dataset and the final Power BI dashboard, where the findings were presented through interactive visualizations.

The results obtained from the SQL analysis were used as supporting evidence for identifying key business insights and understanding the factors that can affect food delivery performance.

,9. SQL ANALYSIS RESULTS

The SQL business analysis provided useful insights into the factors associated with food delivery time. The analysis was performed using the cleaned dataset stored in MySQL.

9.1 Traffic Density

Traffic density showed a clear relationship with delivery time.

Deliveries under Jam traffic recorded the highest average delivery time, while Low traffic recorded a considerably lower average delivery time.

This indicates that road traffic is an important operational factor affecting food delivery performance.

9.2 Weather Conditions

Weather conditions also showed noticeable differences in average delivery time. Cloudy and Fog conditions recorded higher average delivery times compared with Sunny conditions.

This suggests that unfavorable weather conditions can contribute to longer delivery durations.

9.3 Vehicle Type

Vehicle analysis showed differences between vehicle categories. Motorcycles recorded a higher average delivery time compared with Scooters and Electric Scooters.

This comparison provides useful information for understanding delivery performance across different vehicle types.

9.4 Order Type

The average delivery time across Meal, Snack, Buffet, and Drinks orders was very similar, with only small differences between the categories.

Therefore, order type did not show a major difference in delivery duration within this dataset.

9.5 City

City-level analysis showed that Semi-Urban deliveries had a much higher average delivery time. However, the Semi-Urban category contained only a very small number of orders compared with Metropolitan and Urban categories.

Therefore, the Semi-Urban result should be interpreted carefully and should not be generalized as an overall operational trend.

9.6 Festival

Orders during festival conditions showed a higher average delivery time compared with non-festival orders. This indicates that festival periods may be associated with increased delivery duration.

9.7 Multiple Deliveries

The number of multiple deliveries showed a strong difference in average delivery time. Orders involving a higher number of multiple deliveries recorded substantially higher average delivery times.

This suggests that handling multiple deliveries can be an important operational factor affecting delivery performance.

Overall, the SQL analysis helped identify the major categories that require attention when evaluating food delivery performance.

10. DATA VISUALIZATION

After completing the exploratory analysis and SQL business analysis, data visualization techniques were used to present important patterns in the food delivery dataset in a clear and understandable manner.

Python libraries such as Seaborn and Matplotlib were used to create visualizations based on delivery time and different operational factors.

10.1 Delivery Time Distribution

A histogram was created to understand the distribution of delivery time. The visualization showed that most deliveries were concentrated within the lower delivery-time range, while a smaller number of deliveries extended to higher delivery times.

10.2 Average Delivery Time by Traffic

A bar plot was created to compare average delivery time across different road traffic density categories. The

visualization showed that higher traffic conditions were associated with longer average delivery times.

10.3 Average Delivery Time by Weather

A bar plot was used to compare average delivery time across different weather conditions. Cloudy and Fog conditions showed higher average delivery times compared with Sunny conditions.

10.4 Average Delivery Time by Vehicle

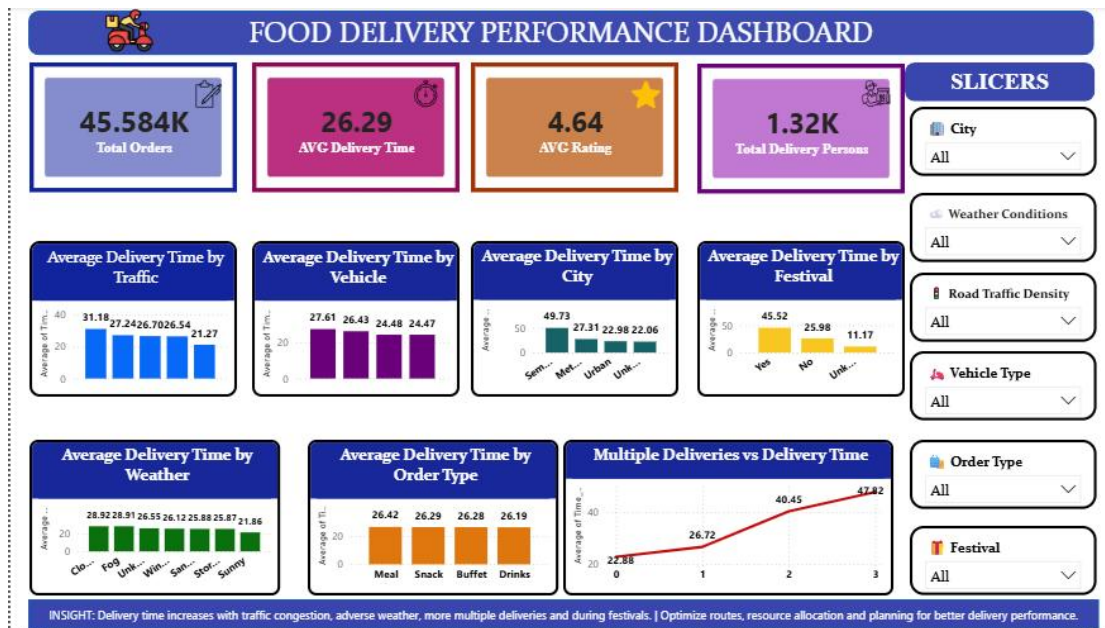
A bar plot was created to compare the average delivery time for different vehicle types. The visualization helped identify differences in delivery performance between motorcycles, scooters, electric scooters, and bicycles.

10.5 Average Delivery Time by Order Type

A bar plot was used to compare average delivery time across Meal, Snack, Buffet, and Drinks orders. The results showed that delivery time was relatively consistent across the different order types.

These visualizations provided a clear graphical representation of the patterns identified during the analysis and helped support the business insights obtained from the dataset.

11. POWER BI DASHBOARD



The final stage of the project was to develop an interactive dashboard using Power BI. The dashboard was designed to present the major findings from the food delivery data analysis in a clear and business-friendly format.

The dashboard is titled Food Delivery Performance Dashboard and focuses on understanding delivery performance and the factors associated with delivery time.

11.1 Key Performance Indicators

The dashboard contains four KPI cards that provide a quick overview of the delivery operation:

- Total Orders
- Average Delivery Time
- Average Rating
- Total Delivery Persons

These KPIs provide users with an immediate summary of the overall delivery performance.

11.2 Main Business Visuals

The dashboard contains visualizations that analyze average delivery time across important business categories.

The major visuals include:

- Average Delivery Time by Traffic
- Average Delivery Time by Weather
- Average Delivery Time by Vehicle
- Average Delivery Time by Order Type
 - Average Delivery Time by City
 - Average Delivery Time by Festival
- Multiple Deliveries vs Delivery Time

These visuals help users compare delivery performance across different operational conditions.

11.3 Interactive Slicers

Interactive slicers were added to allow users to filter the dashboard based on different categories.

The available slicers include:

- City**
- Weather**
- Traffic**
- Vehicle**
- Order Type**
- Festival**

The slicers make the dashboard interactive and allow users to analyze delivery performance for specific categories.

11.4 Dashboard Purpose

The main purpose of the Power BI dashboard is to convert the results of the data analysis into an easy-to-understand visual

report. It allows users to quickly identify patterns, compare delivery performance, and explore the factors that may contribute to longer delivery times.

The dashboard provides a single interactive view of the major findings from the Food Delivery Data Analysis project.

12. KEY BUSINESS INSIGHTS

The analysis of the food delivery dataset provided several important business insights related to delivery performance.

The findings were obtained from the cleaned dataset, SQL business analysis, exploratory analysis, and Power BI dashboard.

Traffic Density: Traffic conditions showed a strong difference in average delivery time. Jam traffic recorded the highest average delivery time, while Low traffic recorded a considerably lower average delivery time. This indicates that road congestion is an important factor affecting delivery performance.

Weather Conditions: Weather conditions also showed differences in delivery time. Cloudy and Fog conditions recorded higher average delivery times, while Sunny conditions recorded a lower average delivery time. This suggests that weather conditions can influence delivery operations.

Vehicle Type: Different vehicle types showed differences in average delivery time. Motorcycles recorded a higher average

delivery time compared with scooters and electric scooters.

This provides useful information for evaluating delivery performance across vehicle categories.

Order Type: The average delivery time across Meal, Snack, Buffet, and Drinks orders was very similar. This indicates that order type does not create a major difference in delivery duration within this dataset.

City: Semi-Urban deliveries recorded a high average delivery time. However, the number of Semi-Urban orders was very small compared with Metropolitan and Urban orders. Therefore, this result should be interpreted carefully and should not be treated as a general operational trend.

Festival Conditions: Festival orders showed a higher average delivery time compared with non-festival orders. This indicates that festival periods may create additional delivery challenges.

Multiple Deliveries: Multiple deliveries showed a significant relationship with delivery time. As the number of multiple deliveries increased, average delivery time also increased

considerably. This indicates that multiple deliveries can be an important operational factor affecting delivery performance.

Overall, the analysis indicates that traffic conditions, weather conditions, festival conditions, and multiple deliveries are important factors to consider when evaluating food delivery performance. Vehicle type also shows differences, while order type has relatively little impact on delivery duration.

13. CONCLUSION

The Food Delivery Data Analysis project successfully demonstrates an end-to-end data analytics workflow, starting from raw data preparation and continuing through data cleaning, exploratory analysis, SQL business analysis, data visualization, and Power BI dashboard development.

The dataset containing 45,584 records and 20 columns was cleaned and validated using Python and Pandas before being imported into MySQL for business analysis. SQL queries were used to analyze delivery performance across different operational factors such as traffic, weather, vehicle type, order type, city, festival conditions, and multiple deliveries.

The analysis showed that factors such as traffic conditions, weather conditions, festival conditions, and multiple deliveries can have a noticeable impact on delivery time. Vehicle type also showed differences in delivery performance, while order type showed relatively consistent delivery times across categories.

The final Power BI dashboard brings these findings together through KPI cards, business visuals, and interactive slicers. This provides a clear and interactive way to understand food delivery performance and explore the factors associated with delivery time.

Overall, the project demonstrates practical skills in Python, Pandas, SQL, data visualization, MySQL, and Power BI, and shows how raw business data can be transformed into meaningful insights for decision-making.

14. Project Files

All project assets are properly documented and available for review:

Zomato Dataset.csv: Raw data file.

food_delivery_cleaned.csv: Cleaned dataset.

food_delivery_pandas.ipynb: Python code for data cleaning and EDA.

Food_Delivery_Analysis.ipynb: SQL queries and execution.

food_delivery_performance_dashboard.pbix: Power BI dashboard file.

GitHubRepository:

[<https://github.com/Dhinakarraaj2006/Food-delivery-data-analysis>]